



FT80 Tall Form Spray Dryer

The Armfield Tall Form Spray Dryer has been specifically designed to enable small quantities of product to be processed. The functional properties of the powders produced are comparable to large-scale production dryer capabilities.

- It can perform operations such as powder spray drying, granulation and micro-encapsulation. Granules and powders of varying sizes can be created for rapid recovery from liquid streams.

PRODUCT CODE: FT80

Categories: Armfield Dairy, Armfield Educational Food Technology, Armfield Flavours And Fragrances, Armfield Ingredients, FT Series - Educational Food Technology, FT80 Tall Form Spray Dryer, IFT Industrial Food Technology

Tags: Bag Filter , Data Logging , Drying , Electric Dryer , Encapsulation , Fluid Bed , FT80 , FT81 , Granulation , Granulator , Micro-Encapsulation' , Microencapsulation , Nozzle , Powders , Spray Chiller , Tall Form Spray Dryer

DESCRIPTION

The Armfield FT80 Tall Form Spray Dryer has been purpose designed to enable laboratory quantities of products to be processed.

Despite the size of the equipment, the powders produced will be comparable to large-scale production dryer capability. This system offers unparalleled flexibility with separate, individually controlled inlet and outlet fans, together with variable nozzle positions and selection of co- or counter-current flow configurations.

This flexibility is enhanced by an unsurpassed range of options, purpose-designed for laboratory and research use.

CO-CURRENT SPRAY-DRYING CONFIGURATION

The liquid to be dried is pumped to the top of the drying chamber by a variable-speed progressing cavity pump, which provides a very smooth flow and constant flow rate.

It is atomised by a two-fluid nozzle and enters the drying chamber as a fine droplet dispersion. Hot air is blown into the chamber, on to the droplets causing flash evaporation of the surface moisture. The bound moisture in the particles is evaporated in the time that they fall to the discharge point at the base of the cone (the residence time). In this configuration the particle temperature will never exceed the wet bulb temperature of the exhaust air.

A variable-speed centrifugal fan provides control over the inlet air flow and a steady inlet temperature is achieved by using a three-term controller in conjunction with an electric heater. A second variable speed fan draws air from the chamber through a cyclone separator. This 'push-pull' fan system provides flexibility to operate the chamber at variable pressures with

variable residence times. Two powder collection points are provided, one at the bottom of the main chamber, and one on the cyclone separator. This enables particles of different sizes to be collected simultaneously and/or separately.

Exhaust-air relative humidity can be measured and controlled so as to enable the system to run at the required RH level. The nozzle is an external-mix two-fluid atomising nozzle. Compressed air is directed onto the liquid stream as it leaves the nozzle body and immediately atomises it. With this type of nozzle the orifice is larger than a single fluid type, and so it is possible to atomise more viscous products and even products containing suspended solids.

TECHNICAL SPECIFICATIONS

GENERAL

Liquid feed: 2-7 l/hr from variable-speed progressive cavity pump

Air flow: Up to 30scfm, using variable-speed fans in 'push-pull' configuration

Chamber residence times: Variable up to 9 seconds

Powder collection points: Separate chamber and cyclone discharge points

Atomisation nozzles: 2-fluid type

SPRAY DRYING

Electric air heater: 4.5kW

Drying air temperature: 50-250°C

Evaporation rates: 1.5-3 l/hr

Typical particle size range: 20-120 microns

FEATURES & BENEFITS

- No clogging, no need for de-blocking devices
- Co-current flow for drying heat sensitive products
- Counter-current flow maximises chamber residence time for non-heat sensitive products
- Relative humidity displayed
- Residence times in the drying cylinder variable up to 9 seconds
- Easy connections to electrical and compressed air utilities
- Data recording to computer for rapid evaluation and comparison
- Available as a Spray Dryer (FT80)
- Drying air temperature up to 250°C
- Maximum flow rate up to 7 l/hr
- Maximum evaporation rate of 3 l/hr
- Tall form chamber straight side to diameter ratio of 3:1, conical discharge section
- Advanced two-fluid nozzle atomisation system
- Supplied with optional nozzle configurations for counter- and co-current atomising
- Individual inlet and exhaust fans with variable-speed control
- Bag filter option
- Flexible controls and full instrumentation provided in the IP65 control console
- Easily dismantled for inspection and cleaning purposes
- Compact, mobile design
- Data logging facility for all key operating parameters
- Powder discharges from two points, chamber and cyclone
- Low noise levels
- Control and measurement of relative humidity

- Spark arrester on input and enhanced safety extraction fan

Configurations

COUNTER-CURRENT SPRAY-DRYING CONFIGURATION

The FT80 Tall Form Spray Dryer is supplied as standard with two nozzles and nozzle hoses, one for co-current and one for counter-current. In counter-current configuration the nozzle is positioned at the bottom of the chamber and sprayed upward into the flow of hot air.

Software

WINDOWS™ DATA LOGGING SYSTEMS

A unique benefit of the Armfield miniature-scale food processing range is that the measured data may be captured and stored to disk. An Armfield FT80-DTA-ALITE Data Logging Accessory transfers results to a PC. Temperatures, pressures and relative humidity can be monitored in real-time.

The data can be displayed in graphical and tabular forms and printed. A PC with a USB port, and running Windows XP or later is required.

ARMSOFT

The armSOFT software enables the operator to select the appropriate stage of the process and a mimic diagram with measured variables is displayed. The speed of the pump can be varied to meet the required flow rate.

Results are saved in a log, which can be viewed and manipulated with the armBUS results viewer. Results can be printed or exported in a spreadsheet format, which can be opened in a wide range of packages for further analysis.

FT80-Spray-Dryer-Software-Screen-Diagram.jpg

Accessories



FLUID BED COOLER ACCESSORY – FT80-60

An optional accessory is available to provide final cooling of the spray-dried powder in a fluidised bed. This feature is particularly useful when drying powders containing fats as the fluid bed helps to crystallise the fat content. Typical applications include infant formulas and whole milk powders.

The FT80-60 Fluid Bed Cooler Accessory

BAG FILTER ACCESSORY – FT80-70

The Bag Filter Accessory is used on the exhaust air outlet. The five micron filter removes virtually all of the remaining product dust in the airstream, resulting in clean air that can be discharged directly into the working area, without requiring further treatment, ducting or extraction fans.

Note: This filter is not designed to extract toxic or hazardous materials.

Applications

- Dairy
- Liquid Foods
- Pharmaceuticals
- Nutraceuticals
- Flavours & Fragrances

Requirements

SINGLE-PHASE ELECTRICAL SUPPLY (SEE ORDERING CODES)

- Compressed air: 1-5 bar pressure, 0.05 m³/min
- PC: Optional (if using the optional Data Logging System a PC with a USB port, and running Windows XP or later is required).



Shipping Specifications

PACKED AND CREATED SHIPPING SPECIFICATIONS

Volume: 2.23m³
Gross weight FT80: 333Kg

Dimensions

Model	FT80
Length:	1.30m
Width:	0.62m
Height:	2.0m

Order Codes

FT80 TALL FORM SPRAY DRYER

FT80-A: Tall Form Spray Dryer: 220-240V/1ph/50Hz*
FT80-G: Tall Form Spray Dryer: 220-240V/1ph/60Hz*
(* Max 35A)

OPTIONAL ACCESSORIES

FT80-60: Fluid Bed Cooler Accessory
FT80-70: Bag Filter Accessory
FT80-DTA-ALITE: Data Logging System

OPTIONAL GMP COMPLIANCE

The FT80 can be produced to full GMP compliance standards. Contact Armfield for further details

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